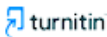


ANNEXURES

ANNEXURE 1: PLAGIARISM REPORT



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ANNEXURE 2: CONFERENCE DETAILS PROCEEDINGS WITH CERTIFICATE PUBLISHED AS THE OUTCOME OF THE PROJECT



ANNEXURE 3: SDG MAPPING

SDG Mapping Table

SDG Goal No.	SDG Title	Relevance Level (Low / Moderate / High)	Justification
SDG 7	Affordable and Clean Energy	High	The proposed SRAM design reduces power consumption through voltage scaling and optimized transistor sizing, making it suitable for energy-efficient and battery-operated systems.
SDG 9	Industry, Innovation and Infrastructure	High	The project contributes to advanced VLSI design by improving SRAM stability and performance without increasing transistor count, supporting innovation in semiconductor memory technologies.
SDG 3	Good Health and Well-Being	Moderate	Energy-efficient and reliable SRAM can be used in biomedical devices and wearable systems, enabling continuous and stable health monitoring applications.
SDG 4	Quality Education	Moderate	The project enhances knowledge in VLSI design, circuit optimization, and simulation tools, contributing to technical education and research development.
SDG 11	Sustainable Cities and	Low	The low-power SRAM design supports embedded and IoT-

SDG Goal No.	SDG Title	Relevance Level (Low / Moderate / High)	Justification
	Communities		based smart systems used in smart city infrastructure.
SDG 12	Responsible Consumption and Production	Low	The design promotes efficient resource utilization by minimizing leakage power and avoiding additional hardware overhead.

ANNEXURE 4: PO PSO MAPPING

PO–PSO Articulation Matrix for the Project

Mapping Level: 1 – Low | 2 – Medium | 3 – High

Program Outcomes (POs) / Program Specific Outcomes (PSOs)	Mapping Level	Justification
PO1 – Engineering Knowledge	3	Applied CMOS fundamentals, SRAM cell design, transistor sizing (CR, PR), and noise margin concepts to develop an optimized 6T SRAM cell.
PO2 – Problem Analysis	3	Analyzed low-voltage SRAM challenges such as reduced SNM, leakage, and process variations through literature review and circuit-level understanding.
PO3 – Design/Development of Solutions	3	Designed a variation-tolerant low-power 6T SRAM cell using transistor sizing optimization without additional assist circuits.
PO4 – Investigation of Problems	3	Conducted simulation-based analysis (SNM, delay, power, leakage) under varying supply voltages and operating conditions to validate the SRAM performance.
PO5 – Engineering Tool Usage	3	Used Cadence Virtuoso and Spectre tools for schematic design, simulation, and performance evaluation of SRAM cell.
PO6 – Engineer and Society	2	Developed energy-efficient memory suitable for IoT and embedded systems, contributing to low-power electronics and sustainable technology.

Program Outcomes (POs) / Program Specific Outcomes (PSOs)	Mapping Level	Justification
PO7 – Ethics	1	Followed ethical research practices, proper citation of literature, and accurate reporting of simulation results.
PO8 – Individual and Team Work	2	Executed the project with guidance, collaboration, and structured task distribution during design and analysis.
PO9 – Communication	2	Prepared detailed technical report, documentation, and explanation of SRAM design, simulations, and results.
PO10 – Project Management	2	Managed design flow including literature review, simulation iterations, result validation, and report preparation within project timeline.
PO11 – Life-long Learning	2	Gained knowledge in advanced SRAM design, low-voltage operation, and variation-aware circuit techniques beyond academic curriculum.
PO12 – Life-long Learning	2	Acquired knowledge in advanced SRAM design, low-voltage operation, and variation-aware circuit techniques beyond the standard curriculum using Cadence tools and research literature.
PSO1 – Electronic Design & IC Systems	3	Designed and analyzed transistor-level SRAM cell, focusing on VLSI circuit design, stability, and performance optimization.
PSO2 – Signal and Image Processing	1	Minimal relevance; limited to general understanding of digital systems where memory is used, no direct implementation.

Program Outcomes (POs) / Program Specific Outcomes (PSOs)	Mapping Level	Justification
PSO3 – Embedded, IoT and Wireless Systems	2	Proposed SRAM design is suitable for low-power embedded and IoT applications requiring energy-efficient on-chip memory.